

Zhongyuan Liang

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EDUCATION

University of California, Berkeley
Ph.D., Computational Precision Health
Advisors: Irene Chen, Ahmed Alaa

Sep 2023 - May 2028 (Expected)
GPA: 4.00/4.00

University of Toronto
B.S. Computer Science, Statistics

Sep 2019 - June 2023
GPA: 4.00/4.00

RESEARCH EXPERIENCE

C.H.E.N. Lab, University of California, Berkeley
Advisor: Irene Chen

Sep 2023 - Present

- Led project revealing latent data bias in treatment modeling due to medication non-adherence; developed pipeline using LLMs to extract adherence information from clinical notes, improving model validity and promoting health equity; work published at CHIL 2025.
- Collaborating with *Samsung Research* team to develop multimodal ICU risk prediction models integrating wearable device data and EHRs, enabling more accurate early warning for critical clinical events; work under review.

Alaa Lab, University of California, Berkeley
Advisor: Ahmed Alaa

Jan 2024 - Present

- Led research in identifying biases in ML-based treatment effect estimation; developed a hybrid meta-learner for robust conditional average treatment effect estimation across diverse observational settings; published at AISTATS 2026.
- Collaborating with *UCSF cardiologists* to develop an LLM agent for adjudicating heart failure with preserved ejection fraction (HFpEF) diagnoses by integrating clinical notes, demonstrating improved performance over current clinical scoring systems.
- Developing a feature-constrained preference learning framework to reduce hallucination in VLMs by regulating internal feature dynamics during DPO training via Modality-Aware Sparse Autoencoders.

Yu Group, University of California, Berkeley
Advisor: Bin Yu

Jan 2024 - Present

- Worked with a multidisciplinary research team to develop an interpretability method for tree-based models that more reliably identifies signal features and improves stability, enabling more trustworthy and personalized interpretation in high-stakes domains such as healthcare; work under review.

ML and Computational Health Care Group, University of Toronto
Advisor: Rahul G. Krishnan

May 2022 - Jan 2023

- Contributed to developing robust statistical methods for detecting covariate shifts in ML models across settings (e.g., between hospitals), achieving state-of-the-art performance; published at ICLR 2023.

PUBLICATIONS

* denotes equal contribution.

1. **Zhongyuan Liang**, Junhyung Jo, Hyang-Jung Lee, Sang Kyu Kim, Irene Y. Chen. **Ontology-aware Contrastive Learning with Cross-Modal Distillation for ICU Risk Prediction**. Under review, 2026.
2. George Ma, **Zhongyuan Liang**, Irene Y. Chen, Somayeh Sojoudi. **Falsifying Sparse Autoencoder Reasoning Features in Language Models**. In: International Conference on Machine Learning (ICML), 2026.

3. **Zhongyuan Liang**, Lars van der Laan, Ahmed Alaa. **Hybrid Meta-learners for Estimating Heterogeneous Treatment Effects**. In: Conference on Artificial Intelligence and Statistics (AISTATS), 2026.
4. **Zhongyuan Liang***, Zachary T. Rewolinski*, Abhineet Agarwal, Tiffany M. Tang, Bin Yu. **Local MDI+: Local Feature Importances for Tree-Based Models**. In: Transactions on Machine Learning Research (TMLR), 2026.
5. **Zhongyuan Liang**, Arvind Suresh, Irene Y. Chen. **Revealing Treatment Non-Adherence Bias in Clinical Machine Learning Using Large Language Models**. In: Conference on Health, Inference, and Learning (CHIL), 2025.
6. Tom Ginsberg, **Zhongyuan Liang**, Rahul G. Krishnan. **A Learning Based Hypothesis Test for Harmful Covariate Shift**. In: International Conference on Learning Representation (ICLR), 2023.
7. Mohi Reza, Angela Zavaleta Bernuy, Emmy Liu, Tong Li, **Zhongyuan Liang**, Calista K Barber, Joseph Jay Williams. **Exam Eustress: Designing a Brief Online Intervention for Helping Students Identify Positive Aspects of Stress**. In: ACM Conference on Human Factors in Computing Systems (CHI), 2023.

ACADEMIC SERVICE

Conference Reviewer

- NeurIPS, ICLR, AISTATS, FAccT, MLHC, ML4H, CHIL 2024 - 2026

Invited Talks

- **Invited Speaker**, *NLP and Language Models in Healthcare*, Health Applications Society (HAS) session, INFORMS Annual Meeting 2025, Atlanta Oct 2025
- (*Upcoming*) **Invited Speaker**, Section on *Statistical Learning and Data Science (SLDS)*, American Statistical Association, Joint Statistical Meetings (JSM) 2025, New York City Nov 2026

Graduate Student Instructor, University of California, Berkeley

- Data102 Data, Inference, and Decisions Fall 2025

Teaching Assistant, University of Toronto

- CSC236 Introduction to the Theory of Computation Fall 2021
- CSC263 Data Structure & Analysis Winter 2022
- MAT135 Calculus I Fall 2020, Fall 2022
- MAT136 Calculus II Winter 2021, Fall 2021, Winter 2022
- PHL245 Modern Symbolic Logic Winter 2023, Winter 2022, Fall 2022

SCHOLARSHIP & AWARDS

- Department of Computer Science Research Award 2021 - 2022
Awarded to fund a summer research internship in Computer Science, University of Toronto
- Nomination of TATP Teaching Excellence Award 2021 - 2022
Nominated by students for teaching excellence as Teaching Assistant
- Robert Bruce In-Course Scholarship 2021 - 2022
Awarded for top academic achievement at New College, University of Toronto
- New College Council In-Course Scholarship 2020 - 2021
Awarded in recognition of academic excellence at New College, University of Toronto
- Faculty of Arts & Science Alumni & Friends Undergraduate Scholarship 2019 - 2020
Recognized for academic excellence in the Faculty of Arts & Science, University of Toronto
- Dean's List Scholar in the Faculty of Arts & Science 2019 - 2022
Awarded on the basis of GPA, University of Toronto